

$$s_{t+i} \sim P(\cdot \mid s_{t+i-1}, a_{t+i-1}^{\text{prev}}) \, ,$$

$$z_{t+i} = E(\mathcal{O}(s_{t+i})) \, , \qquad (i, z_{t+i}) \in \mathcal{F}_{t,k}$$